Special Relativity: Electromagnetism and Maxwell's Equations

Based on lectures by Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Notes reconstructed from the lecture and organized by LazyingArt LLC with Video2Book.

Chapter 1

Electricity and Magnetism as One Relativistic Field

We have studied how a prescribed electromagnetic field moves a charged particle. The unfinished half of the problem is the field itself: how electric and magnetic fields evolve, and how charge and current act as their sources. Maxwell's equations answer those questions. Before writing them, however, it is worth recovering the physical puzzle that made their relativistic structure unavoidable.

This is a deliberate retreat from formalism. The action principle will return, but not yet. The point of beginning elsewhere is to see why the electric and magnetic fields must be parts of one object before deriving their dynamics from an action.

1.1 The Puzzle in Einstein's First Paragraph

During a visit to Belgium, a discussion of the early Solvay conferences led back to Einstein's 1905 paper, *On the Electrodynamics of Moving Bodies*. Its first paragraph states the motivation with remarkable economy. The same electromagnetic phenomenon receives two apparently different explanations in two inertial frames. One observer calls the effect magnetic; another calls it electric. The observed motion agrees, so the two descriptions cannot represent two different pieces of physics.

The paper is now accessible at the level developed in this course. Even if one does not work through every calculation, the opening paragraph is worth reading because it asks the right question before introducing the answer: how can electric and magnetic descriptions change while the physical event does not?

1.1.1 A charge, a magnet, and two descriptions

Take a magnetic field pointing in the y-direction and a charge moving along the x-axis. In the laboratory frame the electric field vanishes, so the force is magnetic:

$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}. \tag{1.1}$$

It points in the remaining transverse direction. The lecture's quick blackboard axes are not meant to fix every right-hand-rule sign; the important facts are that the force is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$, and that reversing the sign of the charge reverses the force.

Question from the audience. Does the force point up or down in the drawing?

Response. That depends on the sign of the charge. Once the orientation of the axes and the sign of q are fixed, the right-hand rule settles the direction. For the present argument only the existence of a transverse force matters. [a]

[a] Lecture timestamp: 00:07:13.

Einstein described a conducting wire rather than an isolated particle. Atoms provide an approximately fixed lattice while mobile charge carriers move through it. At the beginning of the twentieth century a current in a wire was also much easier to observe than the trajectory of one free charged particle. The distinction is inessential here: a moving carrier in a magnetic field feels the same Lorentz force.

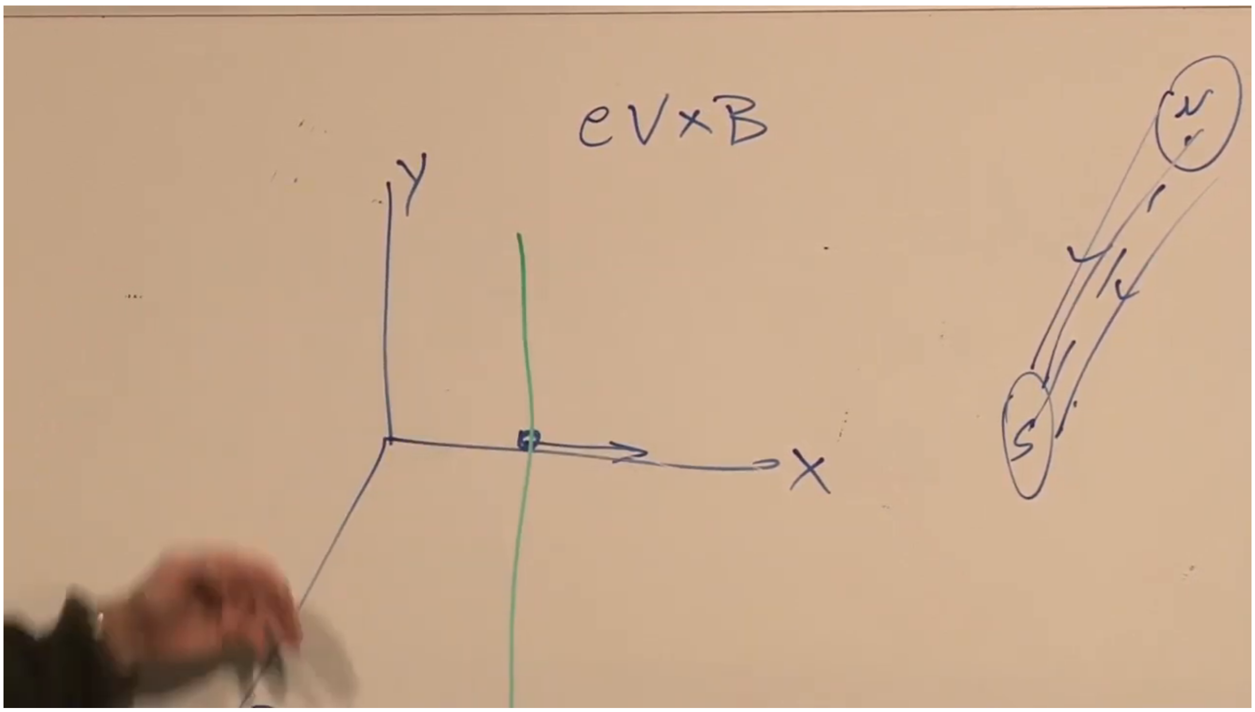

Figure 1.1: The moving charge, transverse magnetic field, and magnet geometry used to restate Einstein's electrodynamic puzzle.

Lecture frame: 00:07:00.

Now keep the wire, or the test charge, fixed and move the magnet instead. The changing magnetic configuration produces an electromotive force, which in local language is an electric field. That electric field pushes the stationary charge in the same transverse direction. Thus the two descriptions are

$$\text{moving charge, fixed magnet:} \quad q\,\mathbf{v}\times\mathbf{B}, \tag{1.2}$$

$$\text{fixed charge, moving magnet:} \quad q\,\mathbf{E}'. \tag{1.3}$$

The relative motion is the same. A theory of relativity must therefore explain how a field that is purely magnetic in one frame acquires an electric component in another.

Electricity and magnetism are not separate substances. They are frame-dependent parts of one relativistic field. Einstein did not yet possess the modern field-tensor notation, and he did not call the required frame changes Lorentz transformations in this paragraph. Nevertheless, the transformation question is already present in full.

1.2 The Electromagnetic Field Tensor

The object that can carry electric and magnetic data together is the antisymmetric tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{1.4}$$

Greek indices run over $0, 1, 2, 3$, with $x^0 = t$; Latin indices run over the three spatial values. Antisymmetry,

$$F_{\mu\nu} = -F_{\nu\mu}, \tag{1.5}$$

sets the diagonal entries to zero and leaves six independent components. Those six are precisely the three components of $\mathbf{E}$ and the three components of $\mathbf{B}$.

With the mostly-plus metric

$$\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1), \tag{1.6}$$

and the convention inherited from the charged-particle action in the previous lecture, define

$$E_m = F_{0m} = \partial_0 A_m - \partial_m A_0, \tag{1.7}$$
$$B_i = (\boldsymbol{\nabla} \times \mathbf{A})_i. \tag{1.8}$$

The lecture corrects a spatial index while writing the first line. The clean relation is the one above: the free spatial index must agree on both sides.

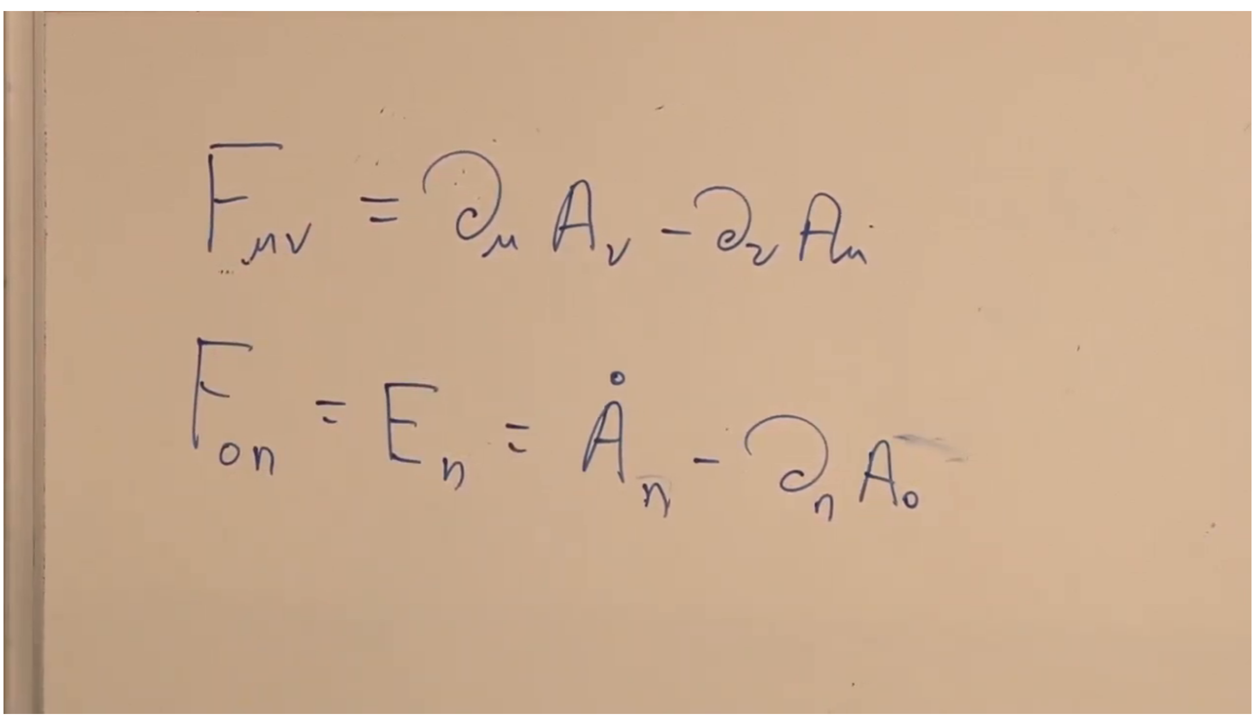

Figure 1.2: The field-tensor definition and the mixed time–space component identified with the electric field.

Lecture frame: 00:13:15.

The magnetic relation is the spatial part of the same definition. For example,

$$B_x = \partial_y A_z - \partial_z A_y = F_{yz}, \tag{1.9}$$

with the other two components obtained by cyclic permutation.

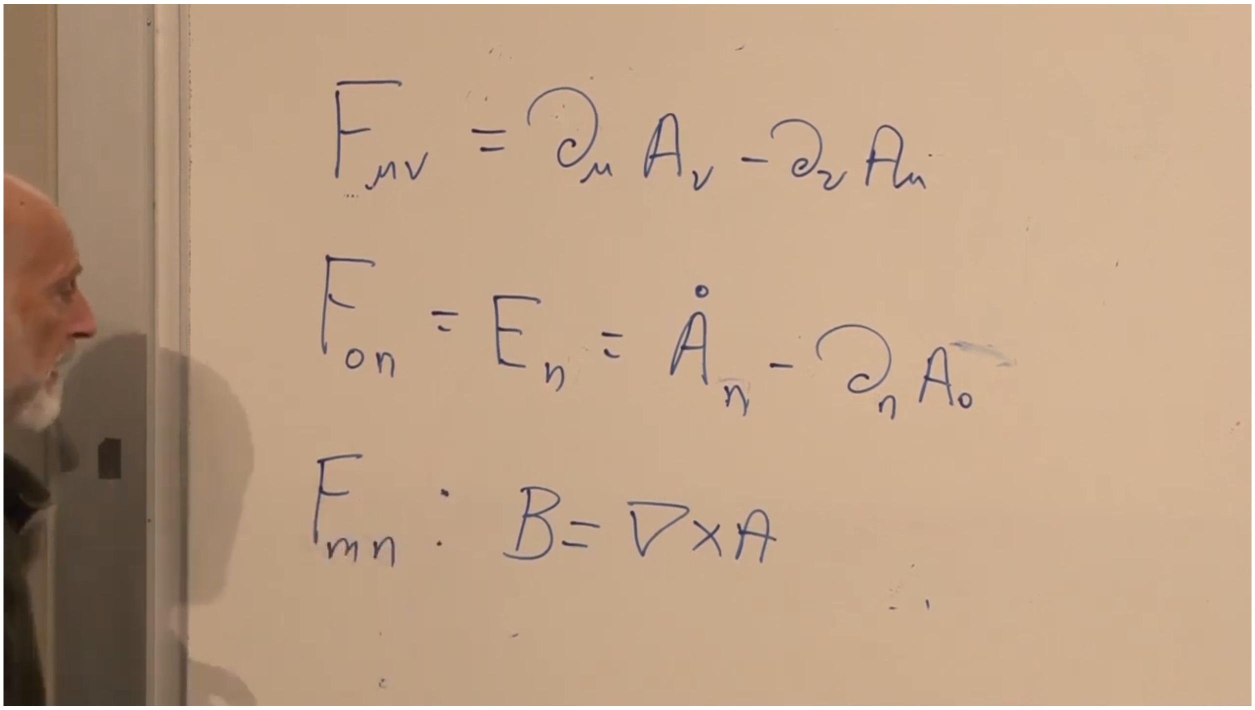

Figure 1.3: The vector-potential formulas for the electric field and the magnetic curl assembled on the board.

Lecture frame: 00:14:00.

1.2.1 The component array

Let the first tensor index label rows and the second label columns, ordered as $0, x, y, z$. Then

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}. \tag{1.10}$$

The mixed time–space edge contains $\mathbf{E}$; the spatial 3×3 block contains $\mathbf{B}$. The lower triangle is not new information, because it is the negative reflection of the upper triangle.

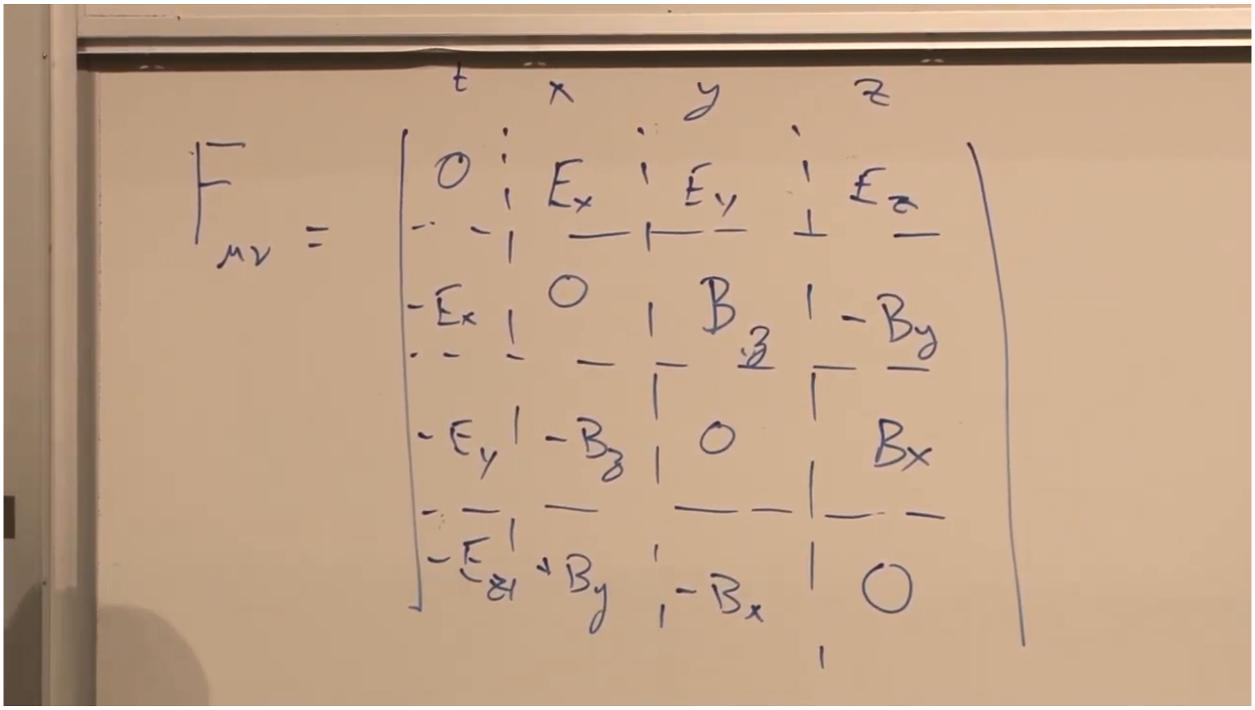

Figure 1.4: The completed lower-index electromagnetic tensor with its electric edge and magnetic spatial block.

Lecture frame: 00:17:15.

Raising both indices changes the sign of every electric entry and leaves the magnetic entries unchanged:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \tag{1.11}$$

An electric component carries one time index and therefore receives one minus sign. A magnetic component carries two spatial indices and receives none.

Question from the audience. Am I reading the matrix correctly that the magnetic components have no time index?

Response. Yes. They occupy the purely spatial block. The electric components are exactly the mixed time–space entries along the first row and first column. [a]

[a] Lecture timestamp: 00:18:55.

1.3 Boosting a Pure Magnetic Field

Consider a boost along x, with $c = 1$:

$$t' = \gamma(t - vx), \qquad x' = \gamma(x - vt), \qquad y' = y, \qquad z' = z, \tag{1.12}$$

where

$$\gamma = \frac{1}{\sqrt{1-v^2}}. \tag{1.13}$$

In matrix form,

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}, \qquad \Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.14}$$

Question from the audience. Is a minus sign missing in the boost matrix?

Response. Yes. The off-diagonal time–space entries carry the minus signs required by $t' = \gamma(t - vx)$ and $x' = \gamma(x - vt)$. The correction is made before the field transformation is used. [a]

[a] Lecture timestamp: 00:23:50.

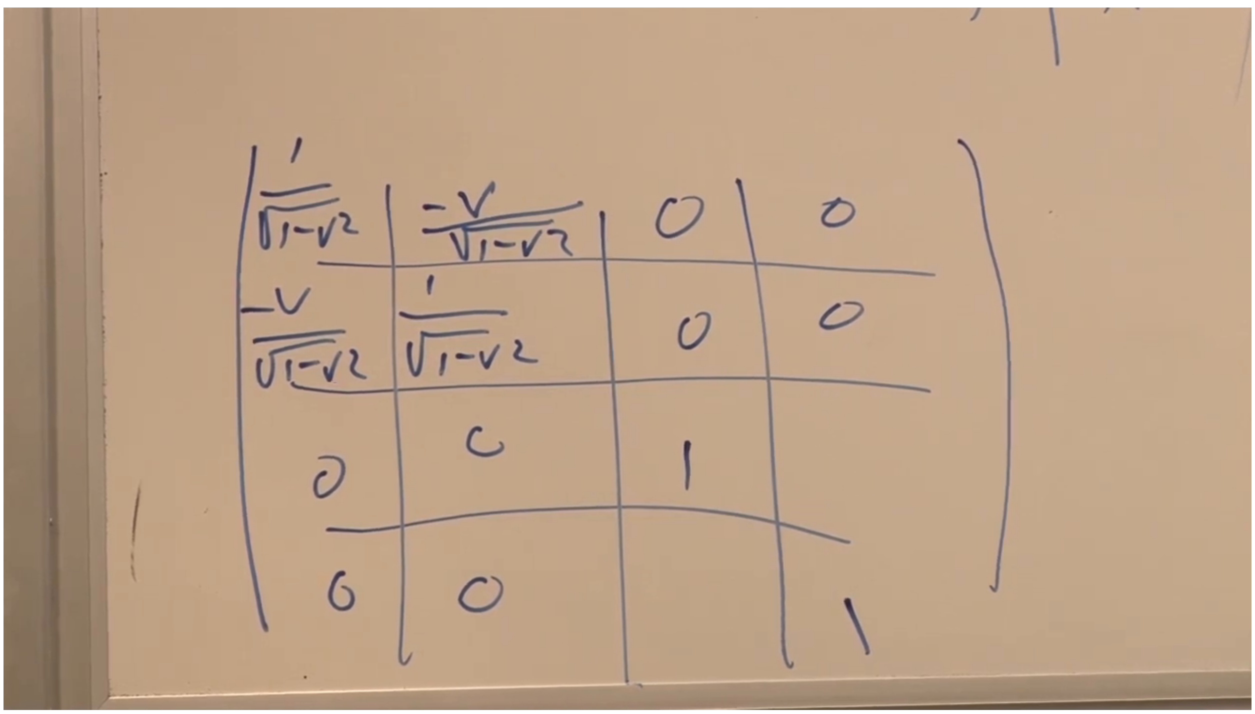

Figure 1.5: The corrected Lorentz-boost matrix for motion along the x-axis.
Lecture frame: 00:24:00.

Every upper tensor index transforms like an upper vector index. Therefore

$$F'^{\mu\nu} = \Lambda^{\mu}{}_{\sigma} \Lambda^{\nu}{}_{\tau} F^{\sigma\tau}. \tag{1.15}$$

The repeated σ and τ are dummy indices. For a tensor with more indices, one transformation matrix would appear for each index.

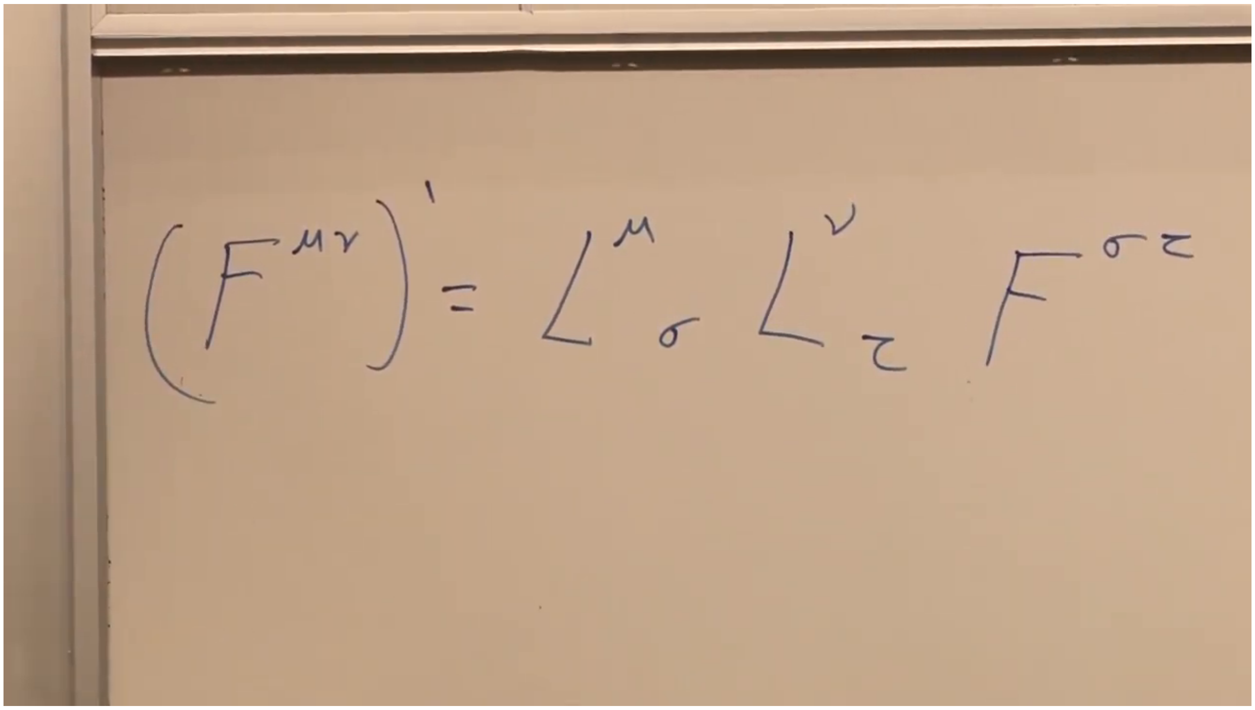

Figure 1.6: The rank-two Lorentz transformation law, with one boost matrix for each field-tensor index.

Lecture frame: 00:26:40.

1.3.1 The component that becomes electric

Assume the unprimed frame contains only a magnetic field B_y. In the conventions of Equation (1.11), the relevant independent component is

$$F^{xz} = -B_y. \tag{1.16}$$

We seek the primed mixed component F'^{0z}, because it carries the primed z-component of the electric field. Equation (1.15) gives

$$F'^{0z} = \Lambda^0{}_\sigma \Lambda^z{}_\tau F^{\sigma\tau} \tag{1.17}$$
$$= \Lambda^0{}_x \Lambda^z{}_z F^{xz} \tag{1.18}$$
$$= (-\gamma v)(1)(-B_y) = \gamma v B_y. \tag{1.19}$$

Since $F'^{0z} = -E'_z$ in this convention,

$$E'_z = -\gamma v B_y. \tag{1.20}$$

The sign depends on the declared axis and field conventions; the invariant point of the calculation is

$$|E'_z| = \gamma |v B_y| \neq 0. \tag{1.21}$$

The double sum collapses because the original field has only this one independent magnetic component.

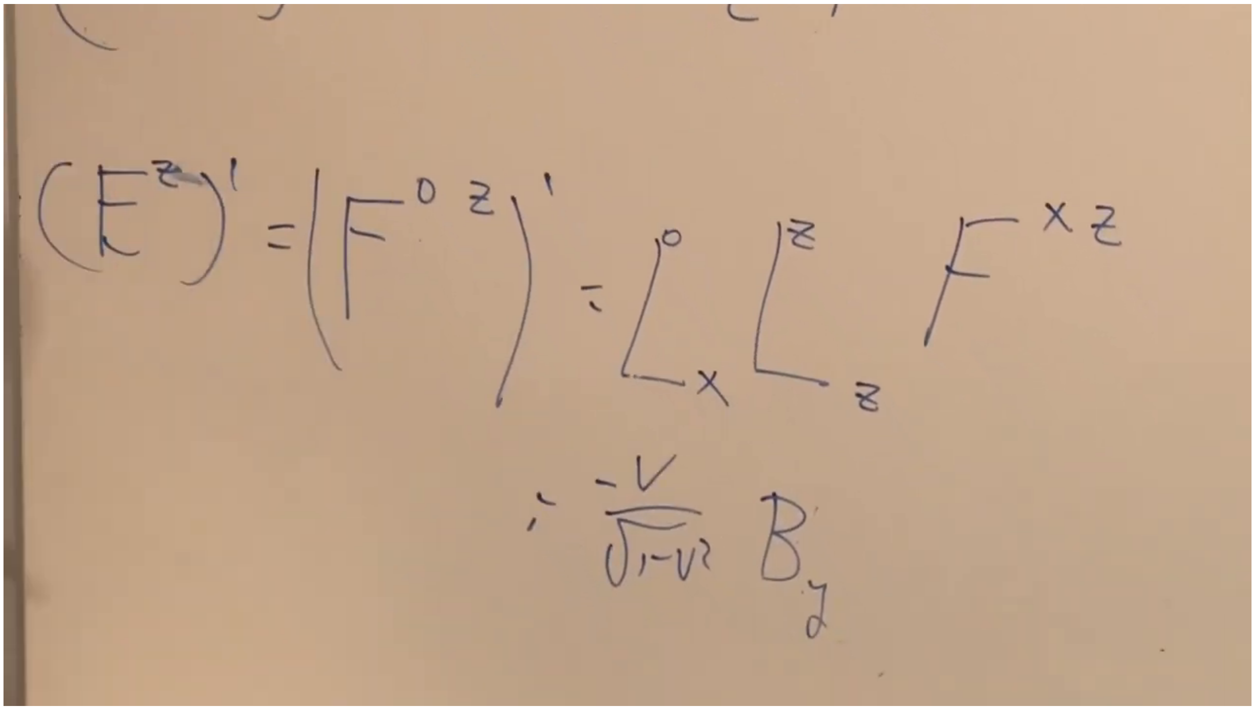

Figure 1.7: The selected tensor component after the boost, showing the factor γv multiplying the original magnetic component.

Lecture frame: 00:32:30.

Choose the primed frame to move with the charge. The charge is then at rest, so there is no magnetic part $q\,\mathbf{v}' \times \mathbf{B}'$ of its force. Yet the force does not disappear: it is $q\mathbf{E}'$. The laboratory observer's magnetic force and the comoving observer's electric force are two descriptions of the same event.

That is the answer to the puzzle in Einstein's first paragraph. The question occupied roughly the first forty minutes of this lecture; the historical joke is that it occupied Einstein for much longer. The calculation is worth the time because it turns the slogan that electric and magnetic fields mix into one explicit tensor component.

1.4 Maxwell's Equations: Identities and Dynamics

Now turn from the force on a charge to the field equations. They will be derived from an action in the next lecture. Here the purpose is first to state them, understand their two different logical roles, and put the identity-based half into covariant form.

Question from the audience. Did we not already derive these equations?

Response. We derived the equation of motion of a charged particle in a prescribed electromagnetic field. Maxwell's equations are equations of motion for the field itself, including the response of the field to charge and current. [a]

[a] Lecture timestamp: 00:35:10.

Question from the audience. How many Maxwell equations are there?

Response. There are eight scalar component equations. Vector notation packages them as two scalar equations and two three-component vector equations, conventionally called four Maxwell equations. [a]

[a]Lecture timestamp: 00:35:24.

We continue to use units in which $c = 1$ and the usual vacuum constants have been absorbed into the definitions of charge and current. Four of the eight component equations follow from the definition $F = dA$. They are kinematic identities. The other four are dynamical and could change if the field action changed.

1.4.1 Three elementary vector operations

For a differentiable three-vector field $\mathbf{A}$,

$$(\boldsymbol{\nabla} \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y, \tag{1.22}$$

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z. \tag{1.23}$$

The remaining curl components follow by cyclically permuting $x \to y \to z \to x$. For a scalar field s,

$$\boldsymbol{\nabla} s = (\partial_x s, \partial_y s, \partial_z s). \tag{1.24}$$

Thus a curl is a vector, a divergence is a scalar, and a gradient is a vector.

Question from the audience. Does anyone need the curl and divergence definitions reviewed in full?

Response. The component definitions are recorded, but the other components need not all be expanded. Cycling the coordinate labels produces them. [a]

[a]Lecture timestamp: 00:37:32.

Commutativity of smooth partial derivatives implies two identities:

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0, \tag{1.25}$$

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} s) = \mathbf{0}. \tag{1.26}$$

Question from the audience. Should we write out the full component proof that the divergence of a curl vanishes?

Response. The class declines. Expanding the components produces pairs such as $\partial_x \partial_y A_z$ and $-\partial_y \partial_x A_z$, which cancel because the derivative order may be interchanged. [a]

[a]Lecture timestamp: 00:40:13.

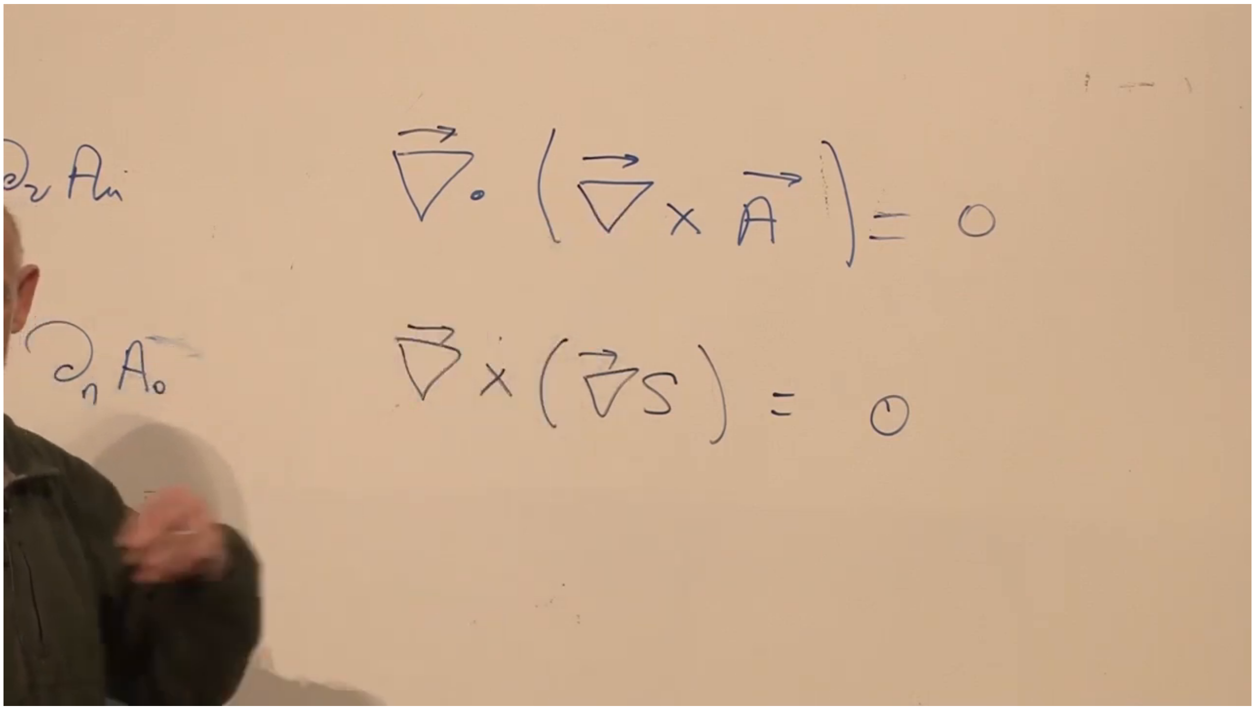

Figure 1.8: The divergence-of-curl and curl-of-gradient identities used to obtain half of Maxwell's equations.

Lecture frame: 00:41:40.

1.4.2 No magnetic source from the vector potential

Taking the divergence of Equation (1.8) gives

$$\boldsymbol{\nabla}\cdot\mathbf{B} = \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) = 0. \tag{1.27}$$

No equation of motion was used. The result follows from the potential description and smooth derivatives alone.

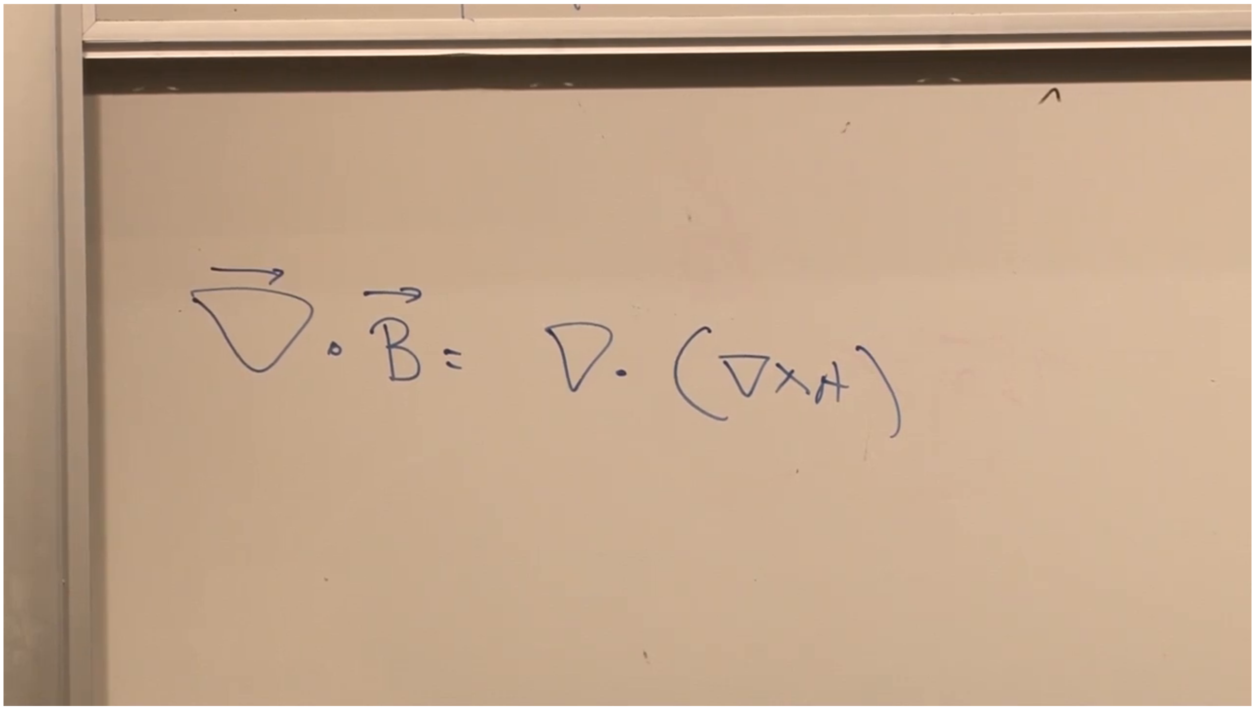

Figure 1.9: The divergence of the magnetic field reduced to the divergence of a curl.
Lecture frame: 00:43:05.

Question from the audience. What is the divergence of the electric field?

Response. It is the electric charge density in the units being used, $\nabla \cdot \mathbf{E} = \rho$. A point charge is the prototype of electric field lines spreading from a source. [a]

[a] Lecture timestamp: 00:43:20.

At this elementary level, Equation (1.27) says there is no magnetic charge density. A smooth, globally defined vector potential cannot produce a magnetic monopole source. Later one must qualify this statement: monopoles can be described with potentials defined on overlapping patches or with singular gauge choices. The lecture postpones that refinement.

1.4.3 Faraday's equation from the electric potential

Equation (1.7) is, in three-vector notation,

$$\mathbf{E} = \frac{\partial \mathbf{A}}{\partial t} - \nabla A_0. \tag{1.28}$$

Although A_0 belongs to a four-vector, it is a scalar under ordinary spatial rotations. Take the curl:

$$\nabla \times \mathbf{E} = \nabla \times \frac{\partial \mathbf{A}}{\partial t} - \nabla \times (\nabla A_0) \tag{1.29}$$

$$= \frac{\partial}{\partial t}(\nabla \times \mathbf{A}) \tag{1.30}$$

$$= \frac{\partial \mathbf{B}}{\partial t}. \tag{1.31}$$

The spatial curl commutes with the time derivative, while the curl of the gradient vanishes.

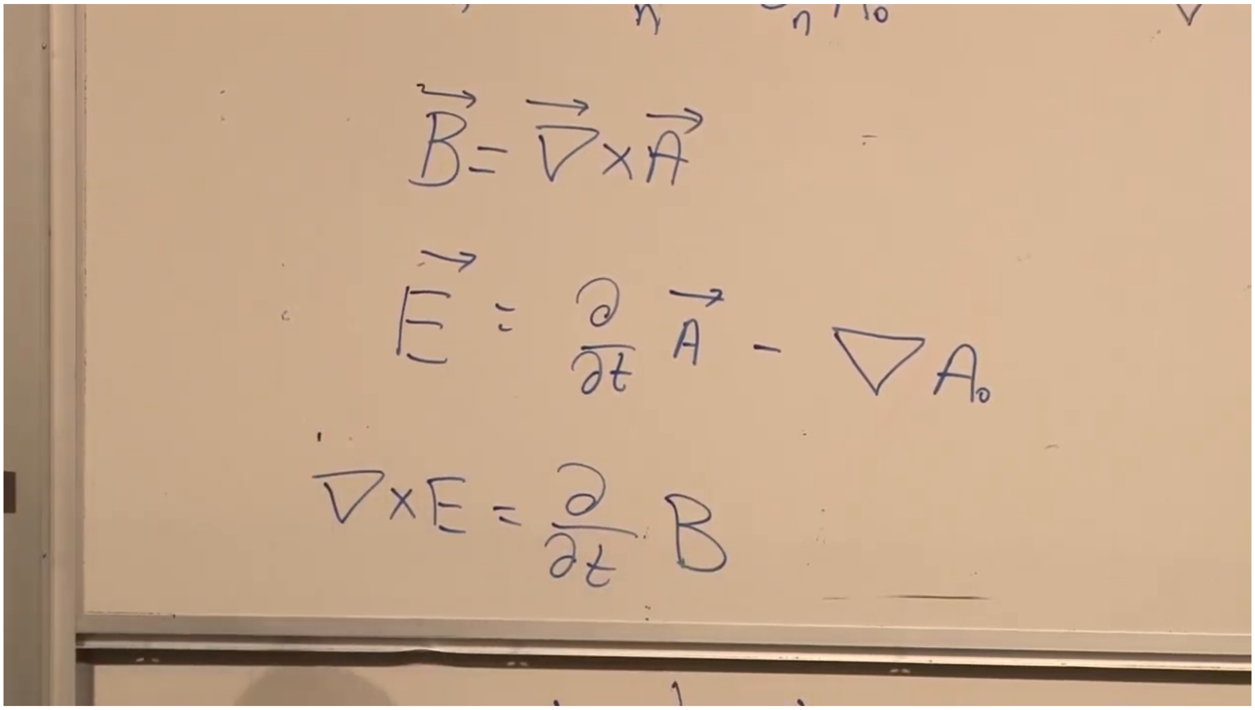

Figure 1.10: The magnetic curl, electric potential formula, and beginning of the curl calculation in one completed board state.

Lecture frame: 00:47:45.

The four homogeneous component equations are therefore

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0, \tag{1.32}$$

$$\boldsymbol{\nabla} \times \mathbf{E} - \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}. \tag{1.33}$$

The first is one scalar equation; the second is a three-component vector equation.

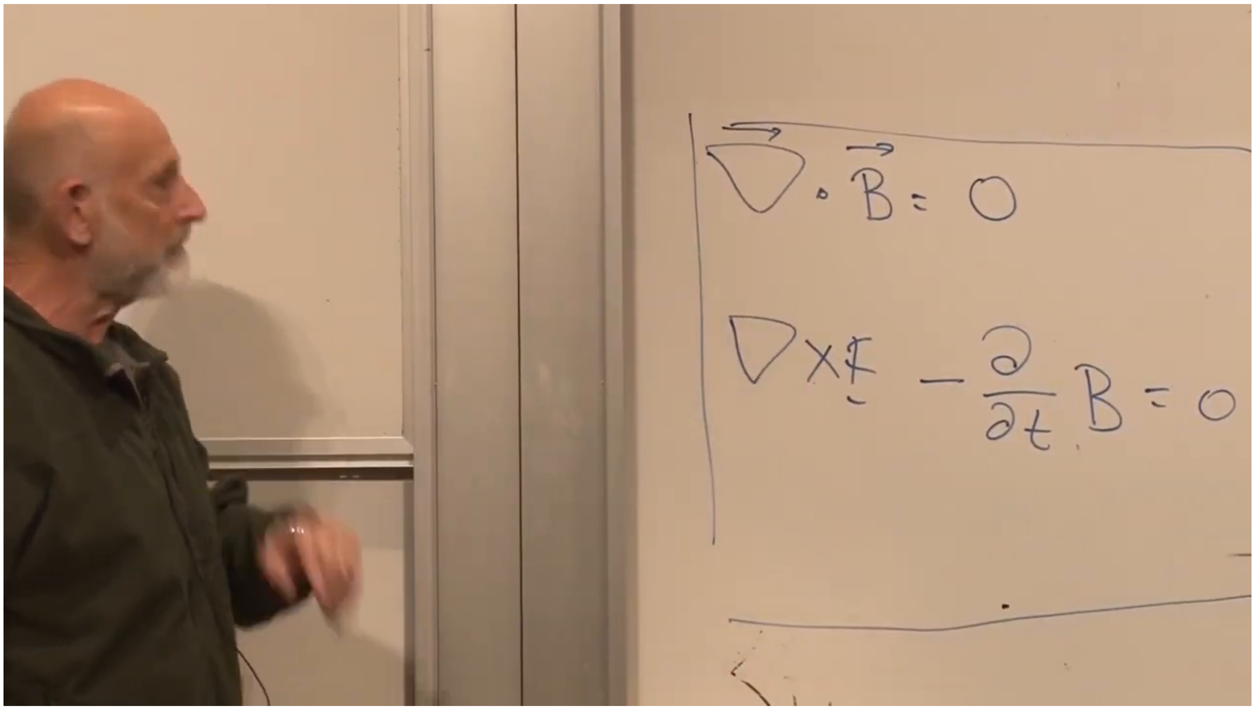

Figure 1.11: The source-free Maxwell pair obtained directly from the potential definitions. *Lecture frame: 00:48:45.*

1.4.4 The dynamical half

The remaining equations are

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho, \tag{1.34}$$

$$\boldsymbol{\nabla} \times \mathbf{B} + \frac{\partial \mathbf{E}}{\partial t} = -\mathbf{J}. \tag{1.35}$$

They are not consequences of $F = dA$. They describe how sources create and change the field, and the next lecture will derive them from the electromagnetic action. The minus sign on the current is fixed by the conventions already chosen: $F_{0i} = E_i$, $F_{ij} = \epsilon_{ijk} B_k$, and $\partial_\mu F^{\mu\nu} = J^\nu$. The live board initially places $\mathbf{J}$ on the right without this minus sign; the resulting mismatch with charge conservation is noticed and corrected in the following lecture.

Question from the audience. What kind of quantity is on the right-hand side of the curl equation?

Response. The left-hand side is a three-vector, so the right-hand side is a three-vector called the electric current density $\mathbf{J}$. The right-hand side of Gauss's equation is the three-dimensional scalar charge density ρ. [a]

[a] Lecture timestamp: 00:50:36.

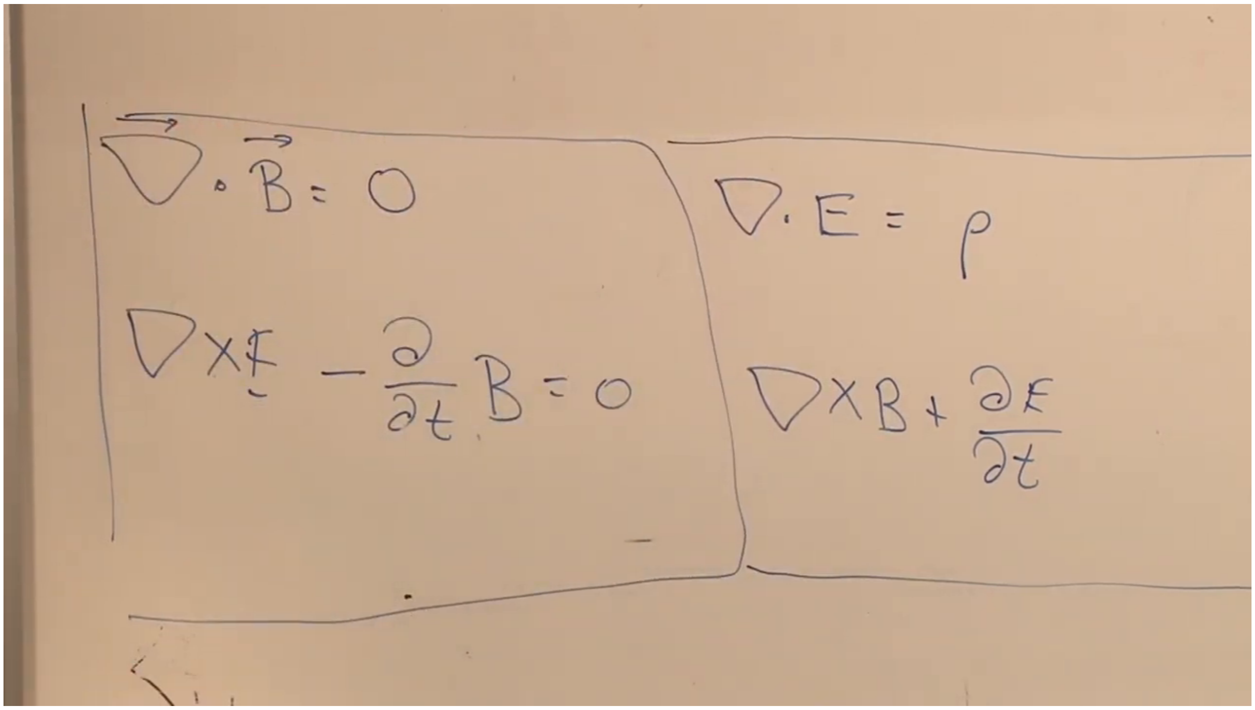

Figure 1.12: The four Maxwell equations grouped into the identity-based pair and the sourced dynamical pair.

Lecture frame: 00:50:40.

1.5 Charge and Current as Spacetime Flow

The symbols ρ and $\mathbf{J}$ must now be given geometric meaning. Charge density is charge per unit volume; current density is charge crossing an oriented area per unit time. The lecture develops both from worldlines so that their later union as a four-vector will not look accidental.

1.5.1 A classroom aside from the neighboring blackboard

Before drawing those worldlines, equations visible in the neighboring room occasion a brief teaching aside. A forthcoming course would introduce quantum mechanics to mathematics faculty who wanted the background needed for quantum field theory and its connections to modern mathematics. The challenge runs in both directions: the physicist must make tacit physical habits explicit, while the mathematician must learn what the formal structures are being asked to describe.

Question from the audience. Who is expected to find that teaching assignment frustrating?

Response. Possibly both sides: physicists teaching an unfamiliar audience and mathematicians confronting the physical habits behind the formalism. The assignment is nevertheless expected to be fun. The class then returns to charge and current. [a]

[a]Lecture timestamp: 00:53:20.

1.5.2 Charge density

Draw time vertically and let the remaining drawn directions stand for three-dimensional space. Charged particles trace worldlines. At one fixed time, place a small spatial cube of volume ΔV across that bundle of worldlines and add the charges of the particles inside it. The charge density is

$$\rho(t, \mathbf{x}) = \lim_{\Delta V \to 0} \frac{\Delta Q}{\Delta V}. \tag{1.36}$$

The two-dimensional square on the board represents a three-dimensional cube; the missing spatial direction is suppressed only because four dimensions cannot be drawn on a flat board.

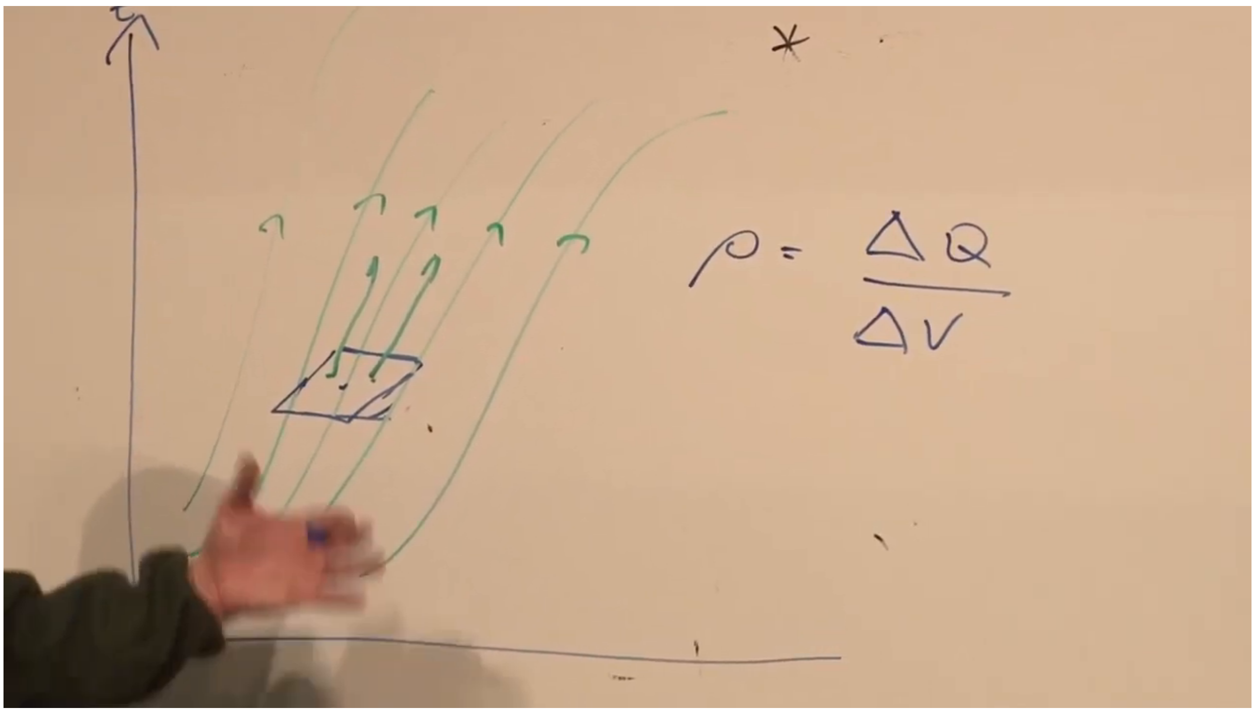

Figure 1.13: Charged-particle worldlines crossing a small spatial cell, with $\rho = \Delta Q/\Delta V$.
Lecture frame: 00:57:20.

1.5.3 Current density

Turn the spacetime cell on its side. To define J_x, choose a small window of area ΔA_x in the yz-plane, perpendicular to the x-axis. Count the net charge that crosses from the negative-x side to the positive-x side during Δt:

$$J_x = \lim_{\Delta A_x, \Delta t \to 0} \frac{\Delta Q_x}{\Delta A_x \, \Delta t}. \tag{1.37}$$

The orientation gives the sign. Windows perpendicular to y and z define J_y and J_z in the same way.

The board struggles honestly with the drawing: a window in the yz-plane looks like a window in the ceiling rather than one in the wall. There are simply more relevant axes than can be displayed at once. The invariant definition is unambiguous even when the sketch is not.

Question from the audience. Is the window in the drawing on the ceiling?

Response. In the room analogy, yes. What matters is that its plane is perpendicular to the x-direction. A yz area measures the x-directed current crossing it. [a]

[a] Lecture timestamp: 00:59:38.

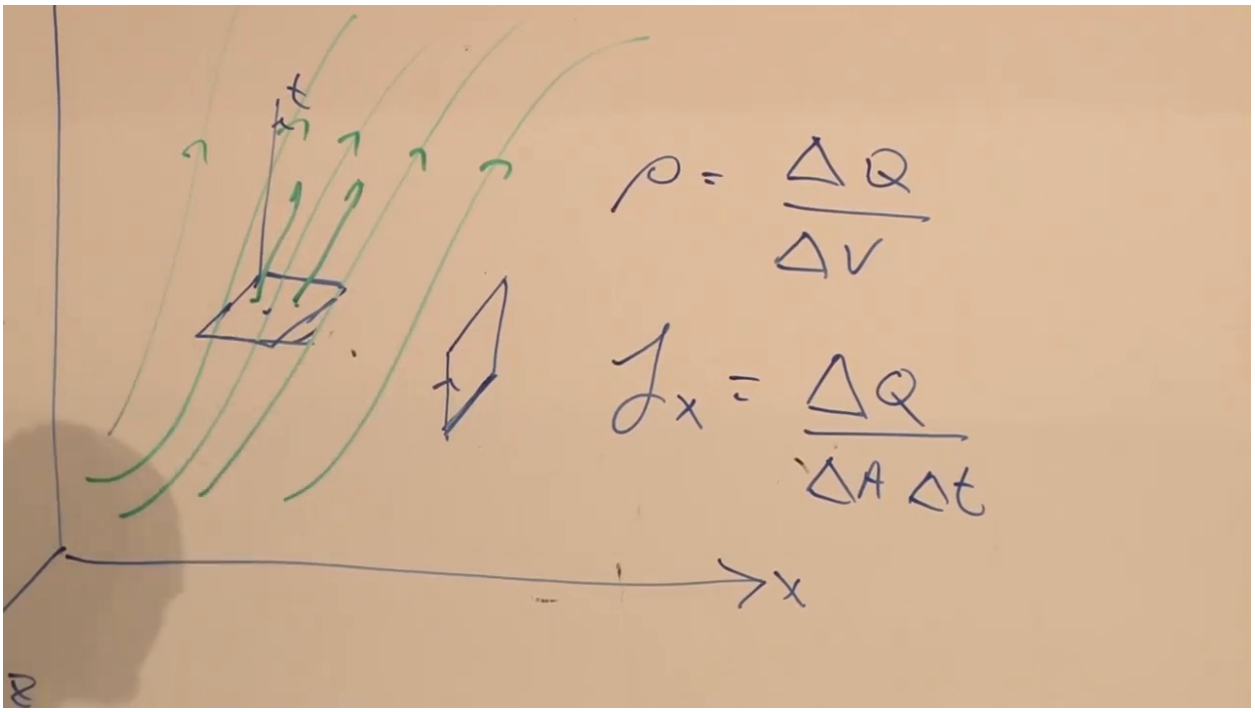

Figure 1.14: The same worldline bundle viewed through a transverse window, giving $J_x = \Delta Q/(\Delta A\,\Delta t)$.

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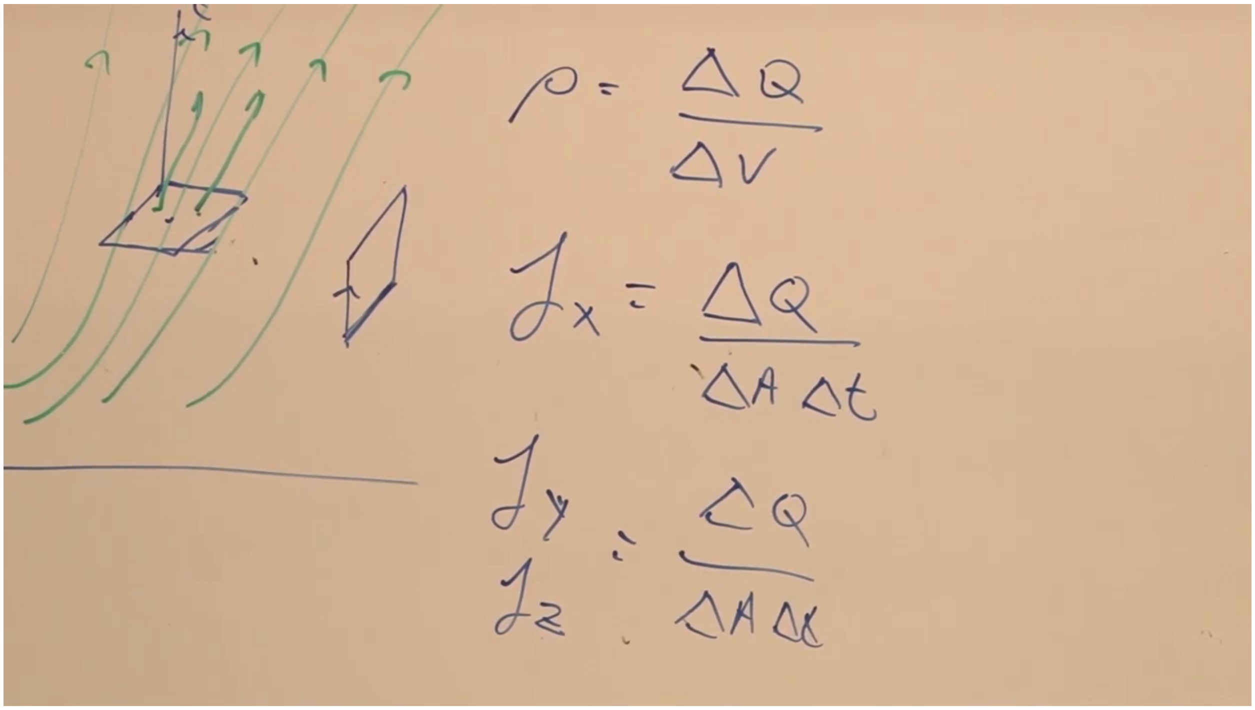

Figure 1.15: The three current components defined by charge crossing windows normal to the three spatial axes.

Lecture frame: 01:02:40.

The four quantities have parallel definitions. A spatial volume at one instant is a hypersurface perpendicular to the time axis, so ρ is the flow of charge through a time-normal window. The J_i are flows through spatially normal windows.

Question from the audience. Are the four numbers one flow in the time direction and three flows in the spatial directions?

Response. That is exactly the useful picture. A spatial volume is a window perpendicular to the time axis; the three transverse areas are windows perpendicular to x, y, and z. Their four fluxes belong together. [a]

[a] Lecture timestamp: 01:04:14.

This suggests the contravariant four-current

$$J^{\mu} = (\rho, J_x, J_y, J_z). \tag{1.38}$$

Question from the audience. Where does the metric enter, and is this current covariant or contravariant?

Response. As defined here, ρ and the three J_i form the components of J^{μ}, an upper-index or contravariant four-vector. The metric lowers it when J_{μ} is required. The covariant Maxwell formulation will show why the upper-index choice is natural. [a]

[a] Lecture timestamp: 01:04:48.

1.6 Interlude at the Break: Signals Near a Black Hole

At the scheduled break, questions turn to a previous discussion of black holes. Although this is a digression from electrodynamics, it is a substantial classroom exchange and illustrates the same discipline of separating descriptions made by different observers.

Question from the audience. If a distant observer sees an infalling object slow near the horizon, should the black hole look covered by everything that has ever fallen onto it?

Response. In Schwarzschild coordinate time, classical infall is seen to approach the horizon asymptotically. One can picture successive matter as increasingly compressed layers near the horizon. Operationally, however, the outgoing signal is increasingly delayed, redshifted, and faint. Light emitted at an ordinary local frequency arrives first as optical light, then infrared, microwave, and radio, with each photon carrying less energy. No realistic distant detector continues to see a bright frozen object forever; it fades into blackness. [a]

[a] Lecture timestamp: 01:06:11.

The infalling observer's clock and locally emitted flashlight remain ordinary to that observer. The distant description differs because the frequency and arrival rate are redshifted. With finite energy and finite bandwidth, there is effectively a last resolvable part of the message.

Question from the audience. Is the object asymptotically approaching the event horizon?

Response. That is the distant Schwarzschild-coordinate description. Quantum mechanically the fading signal must also be distinguished from the black hole's Hawking radiation. At sufficiently late times the weak redshifted signal is observationally lost in the background of the hole's own radiation. [a]

[a] Lecture timestamp: 01:09:24.

Question from the audience. What if the redshifted wavelength grows beyond the scale of the observable universe?

Response. The signal is lost as a practical matter long before that limit. The lecture gives the order-of-magnitude Hawking picture of roughly one typical quantum per light-crossing time,

$$t_{\text{cross}} \sim \frac{R_s}{c}.$$

For a large hole this is extraordinarily weak radiation, and a sufficiently redshifted infalling signal becomes undetectable much earlier than any cosmological wavelength bound. [a]

[a] Lecture timestamp: 01:09:50.

Question from the audience. Does the rate of about one quantum per crossing time stay fixed throughout the life of the black hole?

Response. No. An evaporating Schwarzschild black hole shrinks, so its crossing time decreases and its Hawking temperature rises. The emission therefore accelerates. The dimensional estimate in terms of the instantaneous radius remains useful, but the radius is not constant. Correspondingly, the typical Hawking wavelength shortens as evaporation proceeds. [a]

[a] Lecture timestamp: 01:11:42.

After the break, with the jacket removed, we return to ρ and $\mathbf{J}$.

1.7 Local Conservation of Charge

It is tempting to state charge conservation only as

$$\frac{dQ_{\text{universe}}}{dt} = 0. \tag{1.39}$$

That statement is too weak. Charge might disappear from a laboratory and reappear instantly at Alpha Centauri, or farther away in the Coma Cluster, without changing the total. Such a law would have little experimental bite.

The stronger statement is local: charge inside a region can change only by crossing the boundary of that region. Place a small box around a point. If its charge increases, there must be a net inward current through its six faces; if it decreases, there must be a net outward current.

1.7.1 Flux through opposite faces

Take the two faces perpendicular to x. The inward contribution on one face is evaluated at one value of x; the contribution on the opposite face is evaluated a small distance away. Their difference, divided by the box width, approaches

$$-\partial_x J_x. \tag{1.40}$$

The minus sign appears because positive J_x is defined as flow toward increasing x, while an outward flux decreases the charge in the box. The y- and z-face pairs similarly contribute $-\partial_y J_y$ and $-\partial_z J_z$. Therefore

$$\frac{\partial \rho}{\partial t} = -\partial_x J_x - \partial_y J_y - \partial_z J_z = -\boldsymbol{\nabla} \cdot \mathbf{J}. \tag{1.41}$$

Equivalently,

$$\boxed{\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J} = 0}. \tag{1.42}$$

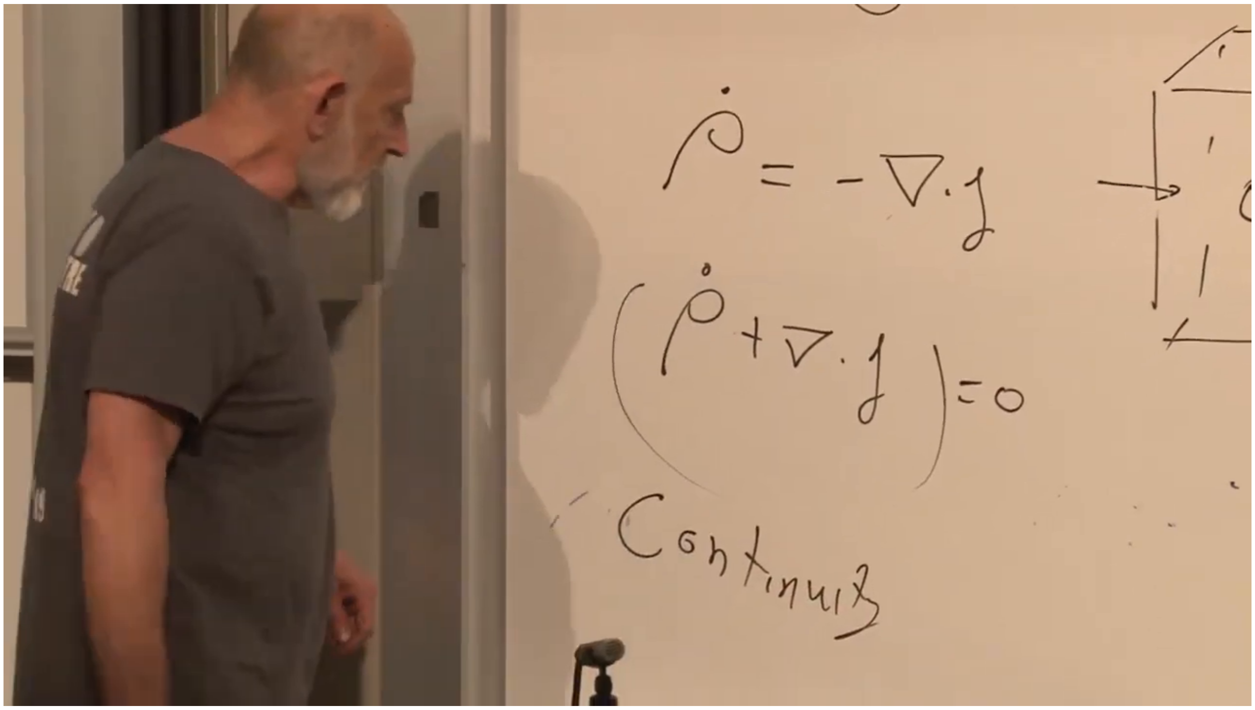

Figure 1.16: The local balance law $\dot{\rho} = -\boldsymbol{\nabla} \cdot \mathbf{J}$, together with the small box whose boundary currents produce it.

Lecture frame: 01:20:45.

This is conventionally called the continuity equation. The lecture prefers the name *local conservation*: continuity can suggest a mathematical smoothness assumption, while the physical content is that charge cannot disappear here and reappear there without a current in between.

1.7.2 Why Maxwell needed the displacement current

Equation (1.42) is not an independent addition to Maxwell's equations. Take a time derivative of Gauss's law and a divergence of the Ampere–Maxwell law:

$$\frac{\partial \rho}{\partial t} = \boldsymbol{\nabla} \cdot \frac{\partial \mathbf{E}}{\partial t}, \tag{1.43}$$

$$-\boldsymbol{\nabla} \cdot \mathbf{J} = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) + \boldsymbol{\nabla} \cdot \frac{\partial \mathbf{E}}{\partial t} \tag{1.44}$$

$$= \boldsymbol{\nabla} \cdot \frac{\partial \mathbf{E}}{\partial t}. \tag{1.45}$$

The two left-hand sides are therefore equal, giving $\partial_t \rho = -\boldsymbol{\nabla} \cdot \mathbf{J}$, exactly Equation (1.42). What matters structurally is that the time derivative of $\mathbf{E}$, the displacement-current term, is essential for the divergence relation to be compatible with charge conservation.

Question from the audience. We have not shown where the two sourced Maxwell equations come from, have we?

Response. No. They are the dynamical target and will be derived from the action next time. The magnetostatic curl law was known before Maxwell; Maxwell's crucial time-dependent addition was the displacement current. Its presence makes the field equation compatible with local charge conservation. [a]

[a] Lecture timestamp: 01:23:15.

The detailed divergence calculation is left as an exercise in the lecture, but its conclusion is not optional: Maxwell's equations require charge conservation.

1.8 The Four-Current and Its Divergence

Write Equation (1.42) in components:

$$\frac{\partial \rho}{\partial t} + \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = 0. \tag{1.46}$$

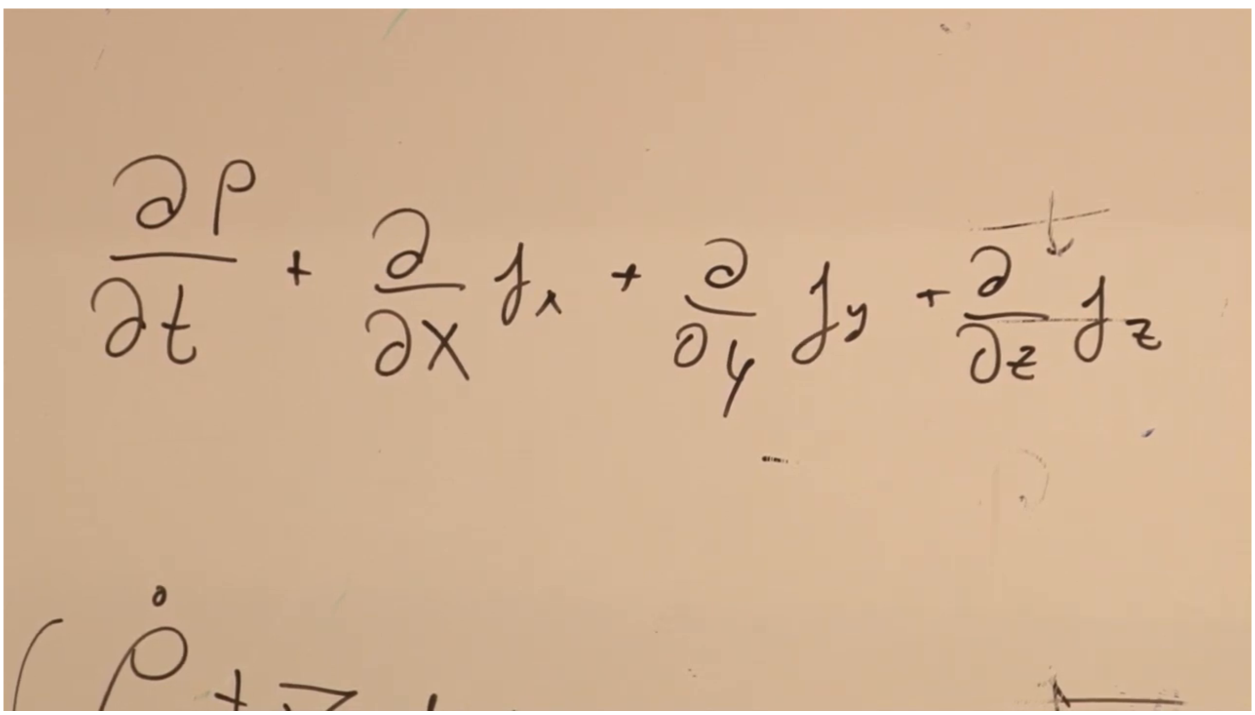

Figure 1.17: Local charge conservation expanded into one temporal and three spatial derivatives. *Lecture frame: 01:25:40.*

With J^μ defined by Equation (1.38), the four terms are one contraction:

$$\boxed{\partial_\mu J^\mu = 0}. \tag{1.47}$$

The lower index on ∂_μ contracts the upper index on J^μ, so the result is a Lorentz scalar. The geometric window construction already suggested this transformation law: ρ and the three current components are four orientations of the same charge flux.

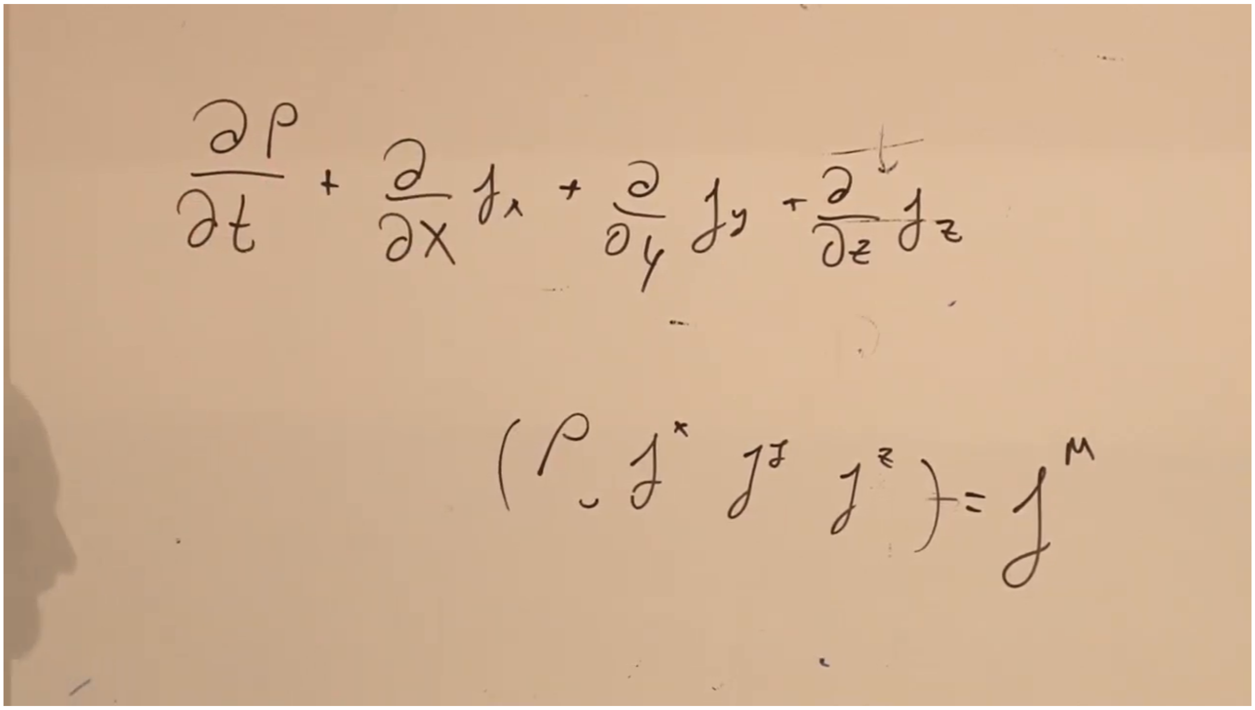

Figure 1.18: The components (ρ, J_x, J_y, J_z) assembled into J^μ and contracted with the four-gradient. *Lecture frame: 01:27:10.*

Question from the audience. When did the time derivative of ρ become a partial derivative?

Response. It was a partial derivative from the beginning. We hold the spatial box fixed and ask how its charge density changes as time changes. We are not following a moving trajectory, so $\dot{\rho}$ here means $\partial\rho/\partial t$ at fixed x, y, z. [a]

[a] Lecture timestamp: 01:28:20.

1.9 The Homogeneous Equations in Covariant Form

We now possess the required objects:

- **E** and **B** are the six independent components of the antisymmetric tensor $F_{\mu\nu}$;
- ρ and **J** are the four components of J^μ;
- local conservation is the scalar equation $\partial_\mu J^\mu = 0$.

The remaining task for this lecture is to write the homogeneous Maxwell pair, Equations (1.32) and (1.33), as one tensor equation.

1.9.1 The cyclic derivative

The answer is

$$\boxed{\partial_\sigma F_{\nu\tau} + \partial_\nu F_{\tau\sigma} + \partial_\tau F_{\sigma\nu} = 0}. \tag{1.48}$$

The three terms cyclically assign the derivative index and the two field-tensor indices. This is often abbreviated as

$$\partial_{[\sigma} F_{\nu\tau]} = 0, \tag{1.49}$$

where brackets denote antisymmetrization.

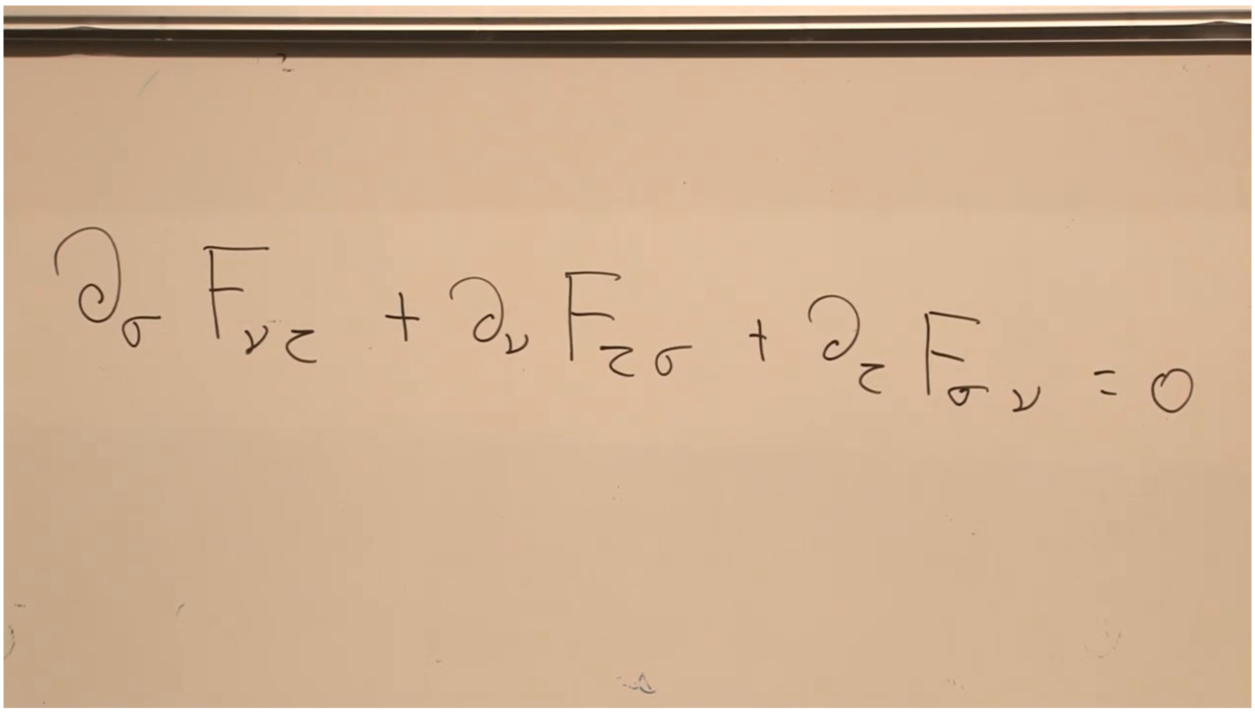

Figure 1.19: The cyclic three-index equation that packages the homogeneous Maxwell equations. *Lecture frame: 01:33:20.*

1.9.2 All three indices spatial

Set

$$(\sigma, \nu, \tau) = (x, y, z). \tag{1.50}$$

Then

$$0 = \partial_x F_{yz} + \partial_y F_{zx} + \partial_z F_{xy} \tag{1.51}$$
$$= \partial_x B_x + \partial_y B_y + \partial_z B_z \tag{1.52}$$
$$= \boldsymbol{\nabla} \cdot \mathbf{B}. \tag{1.53}$$

Thus the all-spatial component of Equation (1.48) is precisely the absence of magnetic divergence.

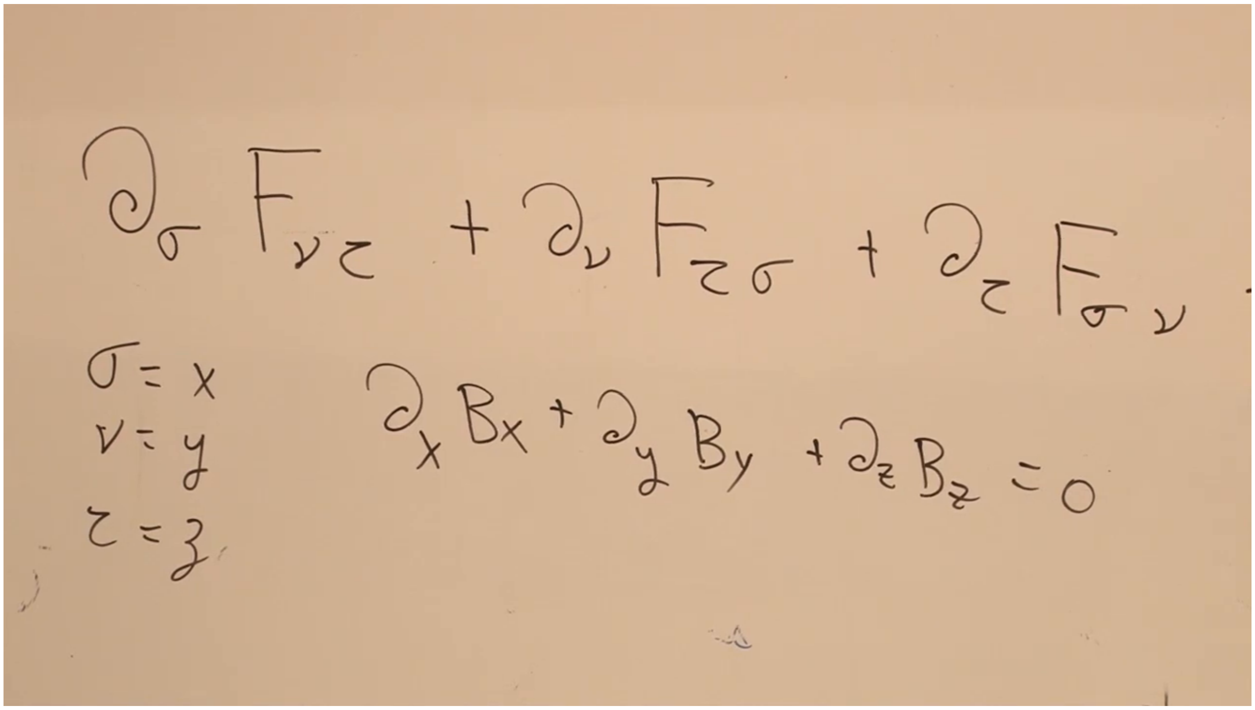

Figure 1.20: The all-spatial component of the cyclic identity reduced to $\partial_x B_x + \partial_y B_y + \partial_z B_z = 0$. *Lecture frame: 01:35:30.*

1.9.3 One time index

Choose, for example,

$$(\sigma, \nu, \tau) = (y, x, 0). \tag{1.54}$$

Then Equation (1.48) combines two spatial derivatives of electric components with a time derivative of a magnetic component:

$$\partial_y F_{x0} + \partial_x F_{0y} + \partial_0 F_{yx} = 0. \tag{1.55}$$

Using the matrix in Equation (1.10), this is one component of

$$\boldsymbol{\nabla} \times \mathbf{E} - \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \tag{1.56}$$

with the relative signs fixed by antisymmetry. Choosing the other pairs of spatial indices supplies the other two curl components.

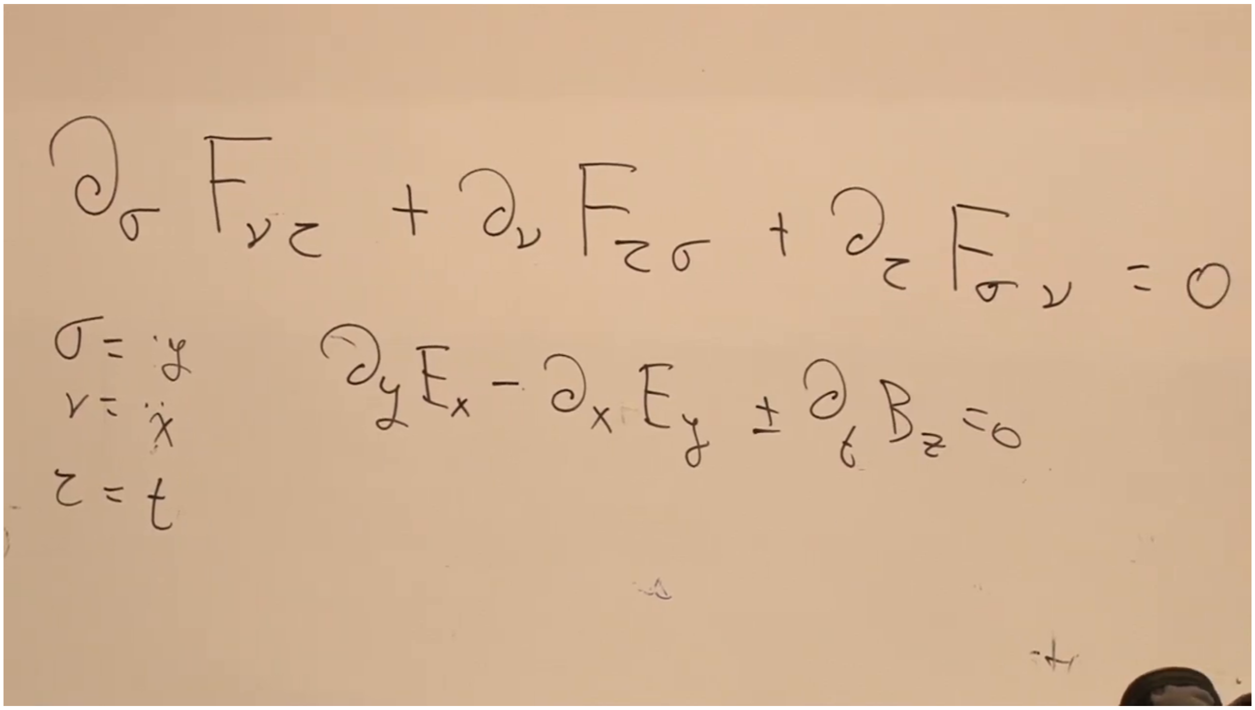

Figure 1.21: A one-time, two-space component check producing one component of Faraday's equation.

Lecture frame: 01:38:20.

There is also a direct proof. Substitute Equation (1.4) into Equation (1.48). Six second-derivative terms appear, and they cancel in pairs because partial derivatives commute. This is the four-dimensional version of the divergence-of-curl and curl-of-gradient identities.

Question from the audience. Is there an h in the name Bianchi?

Response. Yes. The cyclic equation is the electromagnetic Bianchi identity, a special case of a geometric identity that also appears in general relativity. Its deeper geometric interpretation is deferred. [a]

[a] Lecture timestamp: 01:40:28.

1.9.4 Why sixty-four written components contain only four equations

Three indices, each with four possible values, appear to produce $4^3 = 64$ equations. Antisymmetry removes the redundancy:

- if any two indices coincide, the cyclic expression vanishes identically;
- if all three indices are distinct, permutations reproduce the same equation up to sign;
- four dimensions contain only $\binom{4}{3} = 4$ unordered triples of distinct indices.

There are 24 ordered triples with all indices distinct, grouped into four sets of six permutations. The remaining 40 ordered triples vanish. The four independent equations are one magnetic-divergence equation and the three components of Faraday's equation.

Question from the audience. The field tensor has six independent entries, but the three-index expression seems to contain sixty-four equations. Why do only four remain?

Response. Repeated indices give zero, while permutations of three different indices repeat the same relation up to sign. The apparently sixty-four components therefore collapse to four independent equations. [a]

[a]Lecture timestamp: 01:41:58.

Equation (1.48) is manifestly tensorial and therefore Lorentz covariant. In the smooth single-potential description it also says there is neither magnetic charge density nor magnetic current. The dynamical sourced pair is left at exactly the boundary reached in the lecture: its covariant equation and action derivation belong to the next meeting.

1.10 Duality and the Monopole Question

The final questions return to the apparent asymmetry between electric and magnetic sources. In empty space, Maxwell's equations possess the electric–magnetic duality rotation

$$\mathbf{E} \longrightarrow \mathbf{B}, \qquad \mathbf{B} \longrightarrow -\mathbf{E}. \tag{1.57}$$

This is a ninety-degree rotation in an abstract plane with axes $\mathbf{E}$ and $\mathbf{B}$, not a reflection that merely interchanges their labels. With both electric and magnetic sources present, the electric and magnetic four-currents must rotate at the same time.

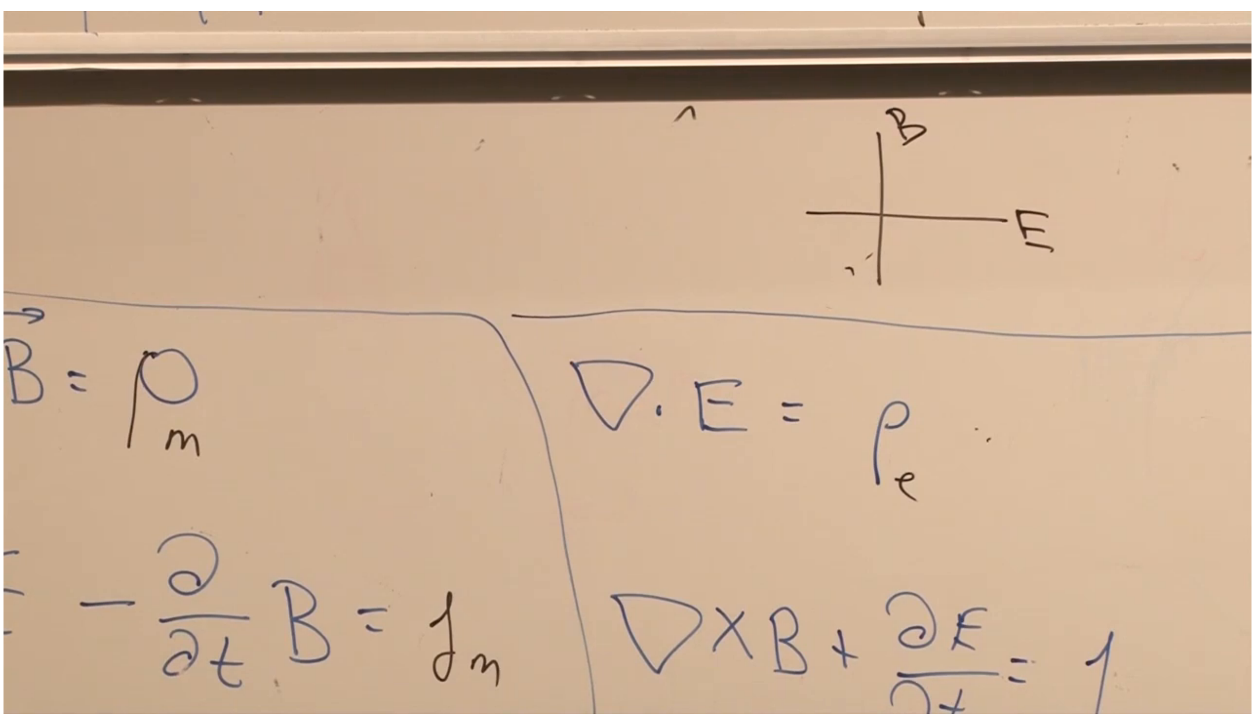

Figure 1.22: The closing electric–magnetic duality picture: a quarter-turn in the E-B plane rather than a simple interchange.

Lecture frame: 01:45:00.

Question from the audience. If source-free equations are dual under electric and magnetic interchange, why are the electric components time–space entries while the magnetic components are space–space entries?

Response. The distinction is frame-dependent within $F_{\mu\nu}$, and the source-free equations admit a duality rotation. The correct operation is $\mathbf{E} \to \mathbf{B}$ and $\mathbf{B} \to -\mathbf{E}$, with source currents rotated similarly if magnetic charge is included. A direct interchange misses the required sign. [a]

[a] Lecture timestamp: 01:43:25.

Question from the audience. Does the clean tensor picture make a magnetic monopole meaningless?

Response. No. It means that a monopole cannot be represented by one smooth vector potential over all of space. The potential must instead be patched, made singular in a gauge-dependent way, or replaced by an equivalent dual description. No elementary magnetic monopole had been observed, and the lecture promises to return to how monopoles fit the formalism rather than forcing the issue at the end of the night. [a]

[a] Lecture timestamp: 01:45:50.

The lecture therefore ends at a precise frontier. Relativity has explained why electric and magnetic fields mix. Potential definitions have supplied the homogeneous Maxwell equations and their Bianchi form. Charge flow has become the four-current, and local conservation has become $\partial_\mu J^\mu = 0$. What remains is dynamical: the electromagnetic action and the covariant sourced field equation.